University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,862: Software Assessment: From Planning to Experimentation

ECTS credits: 6

Overview examination/s

(binding regulations see below)
decentral - Active participation, Analog, Individual work individual grade (20%)
Examination time: Term time
decentral - Portfolio, --, -- (80%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,862,1.00 Software Assessment: From Planning to Experimentation -- English -- Abbad-Andaloussi Amine , Weber Barbara
7,862,2.00 Software Assessment: From Planning to Experimentation: Exercises -- English -- Pandjaitan Adrian

Course information

Prerequisites

The course is open for MCS, MBI and MGM students.
Basic programming and HCI knowledge, as well as solid digital skills.
Interest in startups, product thinking, and willingness to iterate fast.

Learning objectives

- Understand the principles of the Lean Startup methodology and its use of rapid build-measure-learn cycles to reduce uncertainty.
- Understand agile and lean software development practices in the context of iterative product development.
- Identify and analyze user needs using both qualitative and quantitative methods.
- Design and execute experiments to test product hypotheses, including A/B tests and surveys.
- Collect, interpret, and communicate insights from user behavior and physiological measurements (e.g., eye-tracking).
- Apply Lean Startup principles through hands-on project work.

Content

Today's software-driven companies must respond to changing user needs with unprecedented speed. This shift demands new ways of conceiving, developing, testing, and adapting software, moving from traditional upfront planning toward continuous experimentation and iterative delivery. To meet these demands, organizations increasingly adopt Lean Startup principles and Agile development practices, emphasizing rapid build-measure-learn feedback loops as a means to reduce uncertainty and guide product decisions.

This course provides students with the knowledge and practical skills needed to apply the Lean Startup methodology, with a particular focus on the intersection of software engineering and human-computer interaction (HCI). Students will learn how to integrate lightweight experimentation into the software development lifecycle using tools and methods that support fast hypothesis testing, user-centered design, and iterative product improvement.

Structure and indications of the learning and teaching design

The course consists of interactive lectures (2 SWS), labs and seminar parts (2 SWS), and other activities and is based on the pillars below. The in-class exercises are graded within the portfolio grade:

- Lean startup method and hypothesis-driven development.
- Lean software development and agile practices.
- Designing and planning experiments, including ethics of experimentation.
- Rapid user testing: surveys, questionnaires, and A/B Testing.
- Collecting and analysing behavioural and physiological data.
- Qualitative assessment methods.

Literature

- Eric Ries: The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses, Currency; Illustriert. Edition (13. September 2011)
- Mary Poppendieck, Tom Poppendieck: Lean Software Development: An Agile Toolkit for Software Development Managers, Addison-Wesley Professional; 1. Edition (8. Mai 2003)
- Anh Nguyen-Duc, Jürgen Münch, Rafael Prikladnicki, Xiaofeng Wang, Pekka Abrahamsson: *Fundamentals of Software Startups - Essential Engineering and Business Aspects.* Springer 2020, ISBN 978-3-030-35982-9
- Lazar, Jonathan, Jinjuan Heidi Feng, and Harry Hochheiser. *Research methods in human-computer interaction*. Morgan Kaufmann, 2017.
- Kenneth Holmqvist: Eye tracking: a comprehensive guide to methods and measures. Oxford University Press 2011, ISBN 0199697086, pp. I-XIX, 1-537

Session specific literature will be provided throughout the semester via StudyNet.

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Active participation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	20%
Duration	--

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

None

2. Examination sub part (2/2)

Examination modalities

Examination type	Portfolio
Responsible for organisation	decentral
Examination form	Mixed form
Examination mode	--
Time of examination	Term time
Examination execution	--
Examination location	--
Grading type	--
Weighting	80%
Duration	--

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

no regulation necessary

- For written examinations at home (term paper), courses without credits, etc., no specific rules for examination aids are required.
- The regulations of the University of St. Gallen and the rules of academic work (sources and aids must always be identified) are applicable in a subsidiary fashion.
- All written work must be accompanied by a declaration of authorship.

Supplementary aids

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Content

Students will build a portfolio with several build-measure-learn cycles as part of the course. Students will be asked to hand in the portfolio of the different project artifacts, present their project, and reflect on their learning process (up to three presentations (30 % of portfolio grade), a project report and the corresponding project artifact (30 % of portfolio grade), present their project in a final presentation (30 % of portfolio grade) and reflect on their learning process (10 % of portfolio grade)).

As the course goal is to know and apply the lean startup method with its build-measure-learn cycles, the portfolio will consist of multiple build-measure-learn cycles using the Lean Startup approach that must be solved and documented.

Literature

Examination literature includes the course literature, lecture slides, discussion in the classroom, and further materials uploaded to Canvas on the topics listed under course content.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).